GUY IVAN D. OCON

Software & AI Engineer | Digital Transformation | Research & Knowledge Management

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PROFESSIONAL PROFILE

Software and AI engineer with experience building production-oriented applications, data pipelines, and decision-support tools across energy technology, public health, and nonprofit organizations. Combine end-to-end engineering (embedded connectivity, mobile/web apps, cloud infrastructure, and applied AI) with stakeholder consultation, research synthesis, and knowledge management. Experienced translating complex technical and policy information into accessible dashboards, reports, and digital services for cross-sector audiences.

CORE SKILLS

Engineering: Python, JavaScript, SQL, C/C++, React, Vue.js, Flutter, Node.js, PHP, Laravel, REST APIs, Supabase, PostgreSQL, MySQL, Git/GitHub, BLE/ESP32

AI & Data: RAG, LangChain, ChromaDB, LLM fine-tuning, NLP, scikit-learn, TensorFlow, PyTorch, ETL, web scraping, geospatial analytics, time-series forecasting

Research & Delivery: Desk research, SDG indicator analysis, policy-brief development, knowledge management, stakeholder engagement, data visualization, technical documentation, Figma, Power BI

PROFESSIONAL & VOLUNTEER EXPERIENCE

Software Applications Engineer | *Nascent Technologies Corp (Nascent Batteries)* | Jan 2026 - Present

- Architected an ESP32-to-Flutter-to-Supabase battery-monitoring solution, using phones as BLE gateways to eliminate dedicated gateway hardware.
- Develop telemetry acquisition, on-device downsampling, offline buffering, and cloud synchronization; build backend capabilities across PostgreSQL, Auth, Edge Functions, and Realtime.
- Defined phased validation for BLE reliability, background execution, synchronization, alerts, and fault recovery; led internal AI training and designed prompt-engineering and responsible-AI guidance.

Knowledge Management Manager (Volunteer) | *2030 Youth Force Philippines (YFPH)* | 2026 - Present

- Coordinate research and knowledge-management activities supporting the SDG Youth Agenda, including desk research, evidence synthesis, and organization of research outputs.
- Contribute to SDG Youth Agenda progress reporting, SDG indicator assessment, and policy briefs that translate data and research findings into actionable recommendations for stakeholder engagement and policy advocacy.
- Help turn technical findings into accessible, structured knowledge products for youth advocates, nonprofit partners, and decision-makers.

Computer Programmer I to II (Software & AI Engineering) | *Mindanao Center for Disease Watch and Analytics, UP Mindanao* | Aug 2024 - Dec 2025

- Automated Python scraping, geolocation, and ETL workflows for five disease areas, processing 50+ public posts daily and reducing manual processing by approximately 70%.
- Built surveillance data pipelines and analytics modules, including validation and ingestion of 53,000+ dengue and 200,000+ COVID-19 transmission-estimate records.
- Developed at least seven monitoring and analytics modules, plus a RAG-powered public-health assistant, to support data access and disease trend interpretation.
- Collaborated with government and public-health stakeholders to translate reporting requirements into accessible dashboards and decision-support tools.

Student Researcher - RabDash DC | *University of the Philippines Mindanao* | Mar 2023 - Jun 2024

- Prepared and analyzed 400,000+ dog and human rabies records; built data-entry, bulk-upload, and forecasting tools that digitized approximately 80% of Davao City Veterinary Office rabies workflows.

Summer Intern | *UP Diliman - Chemical Engineering Intelligence Learning Laboratory* | Jul 2023 - Sep 2023

- Conducted exploratory analysis of global oil-price datasets and evaluated Gaussian Process Regression for price-trend research.

SELECTED TECHNICAL PROJECTS

ExamGenie | *examgenie.ai* | *React, Vite, Supabase* |

- Shipped a learning platform with flashcards, practice questions, mock exams, and progress analytics for 80+ active users.

BantayLindol PH | *bantaylindolph.netlify.app* | *Python, React, Leaflet, Supabase* |

- Built a PHIVOLCS bulletin ingestion and interactive map analytics workflow covering 20,000+ earthquake records with spatial clustering, filtering, and trend summaries.

ADDITIONAL DIGITAL TRANSFORMATION & CLIENT WORK

Nonprofit Website Modernization | *Philippine nonprofit clients* | Project-based

- Supported nonprofit digital transformation through website redesign, content-management improvements, hosting/migration planning, and more maintainable publishing workflows.

Software Development - International Client Projects | *Luxembourg and Malta-based teams* | Project-based

- Contributed to 2track QCMS, a quality/compliance management application with web interfaces, reporting, data visualization, and PDF generation capabilities.
- Worked on an AI-powered legal-assistance product for a Malta-based team, translating product requirements into application features and usable workflows.

EDUCATION

Master of Engineering in Artificial Intelligence | *University of the Philippines Diliman* | 2026 - Present

BS Computer Science - Cum Laude | *University of the Philippines Mindanao* |

Undergraduate thesis: Web-based Rabies Data Analytics Dashboard for Forecasting Dog and Human Rabies Cases with Vaccine Cost Optimization.